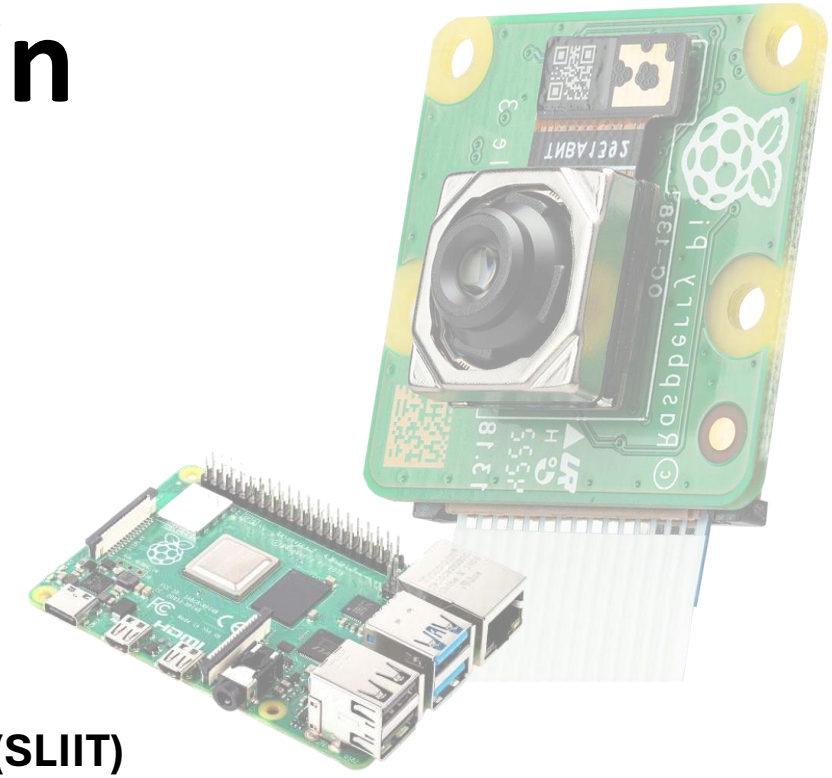
AI-Driven Fabric Defect Detection & Monitoring System for the Apparel Industry in SRI LANKA



Final Presentation

25-26J-452

Faculty of Computing | Software Engineering

Sri Lanka Institute of Information Technology (SLIIT)

AI-Driven Fabric Defect Detection & Monitoring System for the Apparel Industry in Sri Lanka

Panel Comment	Actions Taken
Need to have user testing feedback and need to have recorded demo video for all components and subparts	• Conducted real user evaluation and documented feedback. • Recorded full system demo covering all sub-components. • Demonstrated complete working pipeline (Capture → Anomaly → Quality → Output)
Need to have user testing feedback and need to have recorded demo video for all components and subparts.	• Conducted real user evaluation and documented feedback. • Recorded full system demo covering all sub-components.
Need to have user testing feedback and need to have recorded demo Video for all components and subparts. Need to find a another parameters.	• Conducted real user evaluation and documented feedback. • Recorded full system demo covering the component. • In addition to current parameters, the system now incorporates detection confidence and defect severity from the database.
Need to have user testing feedback and need to have recorded demo video for all components and subparts. Need to validate the data.	• Conducted user testing and documented feedback • Recorded full demo defect detection and localization Process • Validated detected defects and localization outputs against actual fabric conditions.

Problem in Brief

- Manual inspection is slow and error-prone
- Continuous capture causes redundant data & inefficiency
- Poor image quality leads to missed defects
- Lack of real-time processing delays decision-making
- Difficult to accurately locate defects on fabric rolls

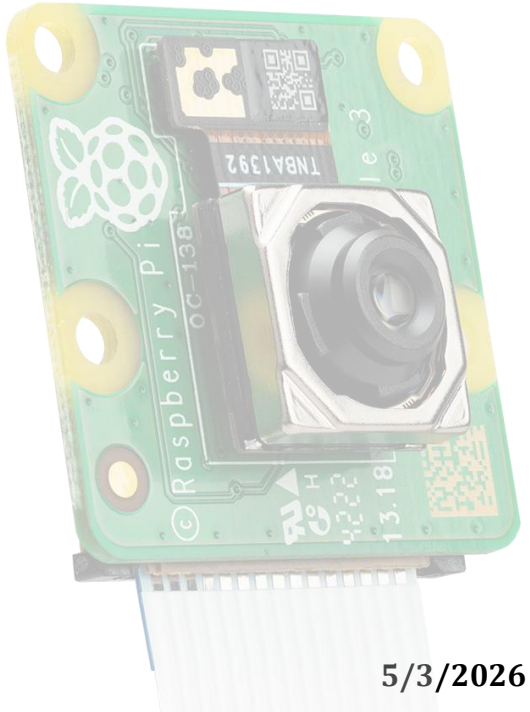
Design Excellence of the Solution

- Intelligent edge-level capture (reduces redundant data)
- Real-time image enhancement for better quality inputs
- AI-based defect detection (~94% accuracy)
- Precise defect localization using IoT (±2 cm)
- Integrated pipeline (capture → enhance → detect → analyze)
- Supports data-driven quality monitoring & prediction

IT22573896 – NAGAHAWATHTHA J.C.D

Edge-Level Intelligent Fabric Capture & Pre-Screening

- Performs real-time fabric frame acquisition (**Raspberry Pi**).
- Uses **encoder-driven event-based capture** to collect frames at fixed spatial intervals, reducing redundant data.
- Detects fabric **presence** and **motion** to ensure only meaningful frames are captured.
- Applies **unsupervised anomaly pre-screening** (Autoencoder + FIS) to identify irregular fabric regions.
- Computes **frame-level quality** (FQS) and filters out low-quality or unusable frames.
- Provides spatially consistent, high-quality frames for downstream defect detection and analytics.



Edge-Level Fog Computing & Image Enhancement

- Performs **real-time fabric image preprocessing at the edge (Raspberry Pi)**
- Identifies **pattern vs non-pattern** and pattern types *(stripe, check, floral, geometric)*
- Applies **adaptive image enhancement** based on image quality and fabric type
- Improves **brightness, contrast, noise reduction, and texture clarity**
- Ensures **consistent, high-quality images for accurate defect detection**
- Reduces **latency and computational load** by processing at the edge

Fabric Defect Detection & Localization

- Performs real-time fabric defect detection using **YOLO-based** deep learning model
- Identifies defects such as holes, stains, cuts and irregularities with bounding boxes
- Accurately localizes defects on the fabric by **mapping them to exact physical positions** in fabric
- Integrates **IoT-based rotary encoder** to track fabric movement and displacement
- Automatically pauses fabric movement **(motor control)** when defects are detected
- Enables easy identification and isolation of defective sections on long fabric rolls

IT22603654 – KAVINDA S.G.D

Predictive Fabric Defect Analytics and Risk Assessment

- Performs **real-time defect prediction** using historical defect data and inspection outputs
- Predicts **next 30 defect positions** along the fabric roll using sequence-based analysis
- Identifies **defect occurrence patterns** such as repeating, drifting, and irregular sequences
- Computes **Roll Risk Score and Supplier Risk Score**
- Applies **machine learning models** to perform Root Cause Analysis (RCA)
- Enables **proactive quality control** by supporting early decision-making and defect prevention

🟧 Results & User Evaluation

- Achieved real-time inspection pipeline (edge + fog)
- High detection accuracy (~94% mAP)
- Precise defect localization (±2 cm)
- Stable real-time performance on edge device

🟩 User Feedback / Acceptance

✓ Reduces manual inspection effort and improves efficiency

✓ Real-time visualization enables quick defect identification

✓ System is easy to use with minimal training

✓ Integrating all outputs into a single display

🟥 Limitations

➢ Performance affected by lighting conditions and camera quality

➢ Fixed thresholds may not adapt to all fabric types

➢ Hardware constraints (Raspberry Pi, 64GB storage)

➢ Limited to fabric width ≤ 90 cm

➢ Difficulty in detecting very small defects

➢ Color detection affected by lighting variations & complex patterns

🟦 Future Improvements

❑ Adaptive thresholding for FIS & FQS (dynamic decision making)

❑ Improved lighting setup and camera calibration

❑ Upgrade to higher-performance edge hardware

❑ Integration with factory systems & cloud analytics

❑ Advanced predictive analytics for defect trends

Research Paper Publication Evidences

8th International Congress on Human-Computer Interaction, Optimization and Robotic Applications

8th International Congress on Human-Computer Interaction, Optimization and Robotic Applications

May 21-23, 2026 | Ankara-Türkiye (Online and In-person)

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ID 255 : AI-Driven Fabric Defect Detection and Monitoring System for the Apparel Industry in Sri Lanka *Author(s) :Janindu Nagahawaththa, Sachintha Kamburugamuwa, Thisara Jayasinghe, Duvidu Gamage, Jenny Krishara, Aparna Jayawardena*	350
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Prof. Dr. Turgut ÖZSEVEN
On behalf of the organization committee
Conference Chair

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Apr 22, 2026

ACCEPTANCE LETTER

Dear Janindu Nagahawaththa [co-author and speaker], **Sachintha Kamburugamuwa** [co-author], **Thisara Jayasinghe** [co-author], **Duvidu Gamage** [co-author], **Jenny Krishara** [co-author], **Aparna Jayawardena** [co-author],

We are glad to inform you that your paper titled; "**AI-Driven Fabric Defect Detection and Monitoring System for the Apparel Industry in Sri Lanka**" has been received, evaluated by reviewers and **ACCEPTED** to be presented **ORAL**, at the "***International Congress on Human-Computer Interaction, Optimization and Robotic Applications***" which will take place in Ankara, Türkiye on May 21, 2026 -May 23, 2026. Thank you for your contribution to the ICHORA2026.

ICHORA2026 is supported by Tokat Gaziosmanpasa University and IEEE Turkey Section.

If you have any questions or need further information, please do not hesitate to contact us, preferably by e-mail, quoting your paper ID: 255 in all correspondence.

Yours sincerely,

Prof. Dr. Turgut ÖZSEVEN
On behalf of the organization committee
Conference Chair

Acceptance Letter

Conference Presentation from May 21-23, 2026

Thank You